

TRIBHUVAN UNIVERSITY
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DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING

A MINOR PROJECT PROPOSAL ON
“DESIGN AND IMPLEMENTATION OF A TWO-WHEELED
SELF-BALANCING ROBOT WITH OPTIMIZED PID CONTROL”

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OUTLINE

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INTRODUCTION

- A self-balancing robot maintains an upright position automatically.
- It is commonly modeled as a two-wheeled inverted pendulum system.
- Uses sensors (e.g., MPU6050) to measure tilt angle and angular velocity.
- A microcontroller processes sensor data continuously.
- A PID controller computes necessary corrections.
- Motors adjust wheel speed and direction based on controller output.
- This keeps the robot balanced around its center of gravity.

OBJECTIVE

- To design and implement a two-wheeled self-balancing robot using an optimized PID control system for stable and efficient balancing.
- To tune and optimize PID parameters for improved stability and response.

LITERATURE REVIEW

Ref.	Author(s)	Key Contribution	Relevance
[1]	Hollinger et al. (2011)	Review of self-balancing robots and stability challenges	Foundational problem domain overview
[2]	Spong et al. (2005)	Robot modeling using Lagrangian & Newtonian mechanics	Basis for inverted-pendulum equations
[3]	Meena & Kumar (2014)	Low-cost PID on a microcontroller-based balancing robot	Applicable to ESP32-based design
[4]	Rahman et al. (2020)	PID strategies for nonlinear inverted-pendulum systems	Guides PID parameter tuning
[5]	Åström & Murray (2008)	Feedback control theory and closed-loop stability	Foundation for PID controller design

LITERATURE REVIEW

Ref.	Author(s)	Key Contribution	Relevance
[6]	Salah & Hossain (2015)	Gyroscope–accelerometer fusion for tilt estimation	Need for reliable sensor fusion
[7]	Madgwick (2010)	Gradient-descent quaternion sensor-fusion algorithm	Lightweight orientation filter for ESP32
[8]	Zhang & Cai (2021)	Embedded balancing robot with IMU and PID	Firmware design and real-time tuning
[9]	Xu et al. (2012)	Simulation-based validation of control strategies	Supports simulation-first prototyping
[10]	Lee & Wong (2011)	Adaptive control for parameter variations	Benchmarks PID vs. adaptive methods

PRINCIPLE DIAGRAM

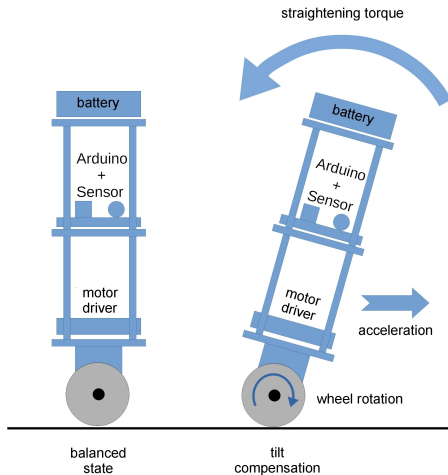


Figure: Tilt Compensation Principle

MATHEMATICAL MODEL: INVERTED PENDULUM

Equation of motion:

$$(I_b + mL^2)\ddot{\theta} - mgL \sin \theta = -\tau \quad (1)$$

$$\text{Linearised (small } \theta): (I_b + mL^2)\ddot{\theta} - mgL \theta = -\tau \quad (2)$$

where,

- m mass of the robot body
- L wheel axis to centre of mass
- I_b body moment of inertia
- g gravitational acceleration
- θ tilt angle (from vertical)
- τ corrective motor torque

Instability insight: With $\tau = 0$, equation (2) becomes $\ddot{\theta} = \frac{mgL}{I_b + mL^2} \theta$, which has an exponential growing solution – the upright pendulum is inherently unstable. A PID controller provides the τ needed to stabilise it.

THEORITICAL BACKGROUND

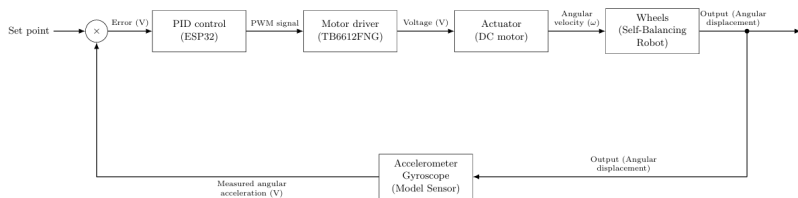


Figure: Closed-loop PID control system of the self-balancing robot

Flow: Setpoint $\rightarrow$ Error $\rightarrow$ PID (ESP32) $\rightarrow$ PWM $\rightarrow$ Motor Driver $\rightarrow$ DC Motors $\rightarrow$ Wheels $\xrightarrow{\text{feedback}}$ IMU Sensor $\rightarrow$ Error

REQUIREMENT ANALYSIS

● **Functional Requirements:**

- The system should maintain balance using an inverted pendulum approach.
- It must continuously measure tilt using an IMU sensor.
- A PID control algorithm should be used for real-time error correction.
- PWM signals must be generated for motor speed and direction control.

● **Hardware Requirements:**

- A microcontroller such as Arduino or ESP32.
- An IMU sensor (accelerometer and gyroscope).
- A motor driver (L298N or TB6612FNG).
- DC motors with wheels.
- A stable power supply.

REQUIREMENT ANALYSIS

- **Software Requirements:**

- Support for real-time sensor data acquisition.
- Implementation of PID control algorithm.
- Generation of PWM signals for motor control.
- Capability for PID parameter tuning and optimization.

- **Non-Functional Requirements:**

- Fast system response.
- High stability and accuracy.
- Resistance to noise and disturbances.
- Reliable operation.
- Compact design and energy efficiency.

SIMULATION AND PROTOTYPING

- The system is first tested in simulation before physical implementation.
- Simulation helps verify circuit connections, sensor interfacing, and PID behavior.
- The virtual model includes MPU6050, ESP32, TB6612FNG motor driver, and DC motors.
- Components are connected according to the proposed hardware design.
- Initial PID gains (K_p , K_i , K_d) are adjusted in simulation.
- Tilt response is observed to evaluate system behavior.
- Validated PID parameters are used as a baseline for real-world calibration.

SYSTEM DESIGN AND HARDWARE IMPLEMENTATION

- The self-balancing robot operates as a closed-loop control system.
- It consists of a microcontroller (Arduino/ESP32), IMU sensor, motor driver, and DC motors.
- The IMU sensor measures acceleration and angular velocity.
- Sensor data is sent to the microcontroller for processing.
- The controller generates control signals to drive the motors via the motor driver.
- The motors adjust movement to maintain balance.
- Proper circuit connections and mechanical design ensure stable system operation.
- The circuit diagram of the system is presented in figure below.

CIRCUIT DIAGRAM OF SELF-BALANCING ROBOT

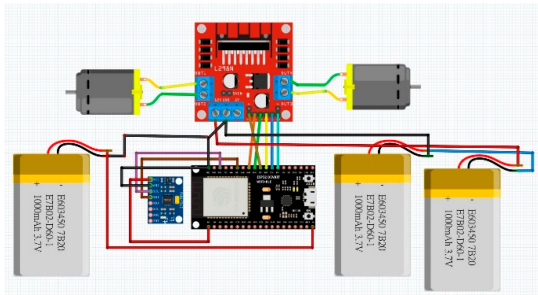


Figure: Circuit diagram of a self-balancing robot

Hardware Components and Their Use

Table: HARDWARE COMPONENTS AND THEIR USE

S.N	Component	Use
1	MPU6050 (IMU Sensor)	Measures tilt and angular motion
2	Microcontroller (ESP32/Arduino)	Processes data and implements PID control
3	Motor Driver (TB6612FNG)	Controls speed and direction of motors
4	DC Gear Motors	Provide motion for balancing
5	Li-ion Battery	Supplies power to the system
6	Buck Converter	Regulates voltage for components
7	Wheels	Enable movement of the robot
8	Chassis	Supports and holds all components
9	Jumper Wires	Provide electrical connections

SENSOR PROCESSING AND CONTROL ALGORITHM

- The IMU sensor continuously measures the robot's tilt and motion.
- Raw sensor data is filtered to obtain a stable and accurate angle.
- The measured angle is compared with the desired upright position.
- The difference generates an error signal.
- The error signal is fed into a PID controller.
- The PID controller computes corrections using proportional, integral, and derivative terms.
- A PWM signal is generated to control motor speed and direction.
- The motors adjust movement to maintain balance.

MOTOR CONTROL AND SYSTEM WORKFLOW

- The controller generates a PWM signal for motor control.
- The motor driver uses this signal to regulate motor speed and direction.
- DC motors respond by adjusting movement accordingly.
- The system operates in a continuous feedback loop.
- Steps include sensing tilt, calculating error, applying PID control, and adjusting motors.
- This continuous process maintains the robot's balance.

CALIBRATION, TESTING, AND PERFORMANCE EVALUATION

- Sensor calibration and PID tuning are performed for stable operation.
- These processes help minimize oscillations in the system.
- The robot is tested under various operating conditions.
- Performance metrics such as response time, accuracy, and stability are evaluated.
- The system is verified for reliable performance against disturbances.

EXPECTED BEHAVIOUR

- Robot maintains **upright position** on two wheels
- Responds quickly to disturbances which minimises oscillations
- Recovers from external tilts efficiently
- Improved stability over non-optimised systems
- Faster response time with optimised gains

QUANTITATIVE TARGET

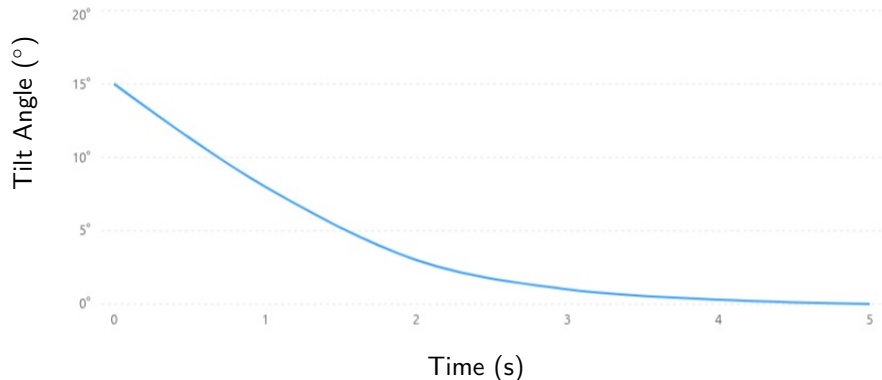


Figure: Expected tilt angle vs. time response of the self-balancing robot

- Initial tilt of **15°** → return to equilibrium within **4–5 seconds** *no overshoot, critically damped* response.

EPILOQUE

S.N	Activity	W1	W2	W3	W4	W5	W6	W7	W8
1	Proposal Report Writing	✓	✓						
2	Component Procurement	✓	✓						
3	Chassis Design & Assembly		✓	✓					
4	Prototyping & Sensor Testing			✓	✓				
5	Basic PID Implementation				✓	✓			
6	PID Tuning & Optimization					✓	✓		
7	Hardware Integration & Wiring						✓		
8	Additional Feature Development						✓	✓	
9	System Integration & Testing						✓	✓	
10	Bug Fixing & Improvements							✓	✓
11	Weekly Progress Reports		✓	✓	✓	✓	✓	✓	
12	Final Report & Presentation Prep							✓	✓

Project Duration: 8 Weeks (2 Months)

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Thank You!

We welcome your questions and suggestions.



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